Contour Integrals: Parametrization, Exactness, Winding, and Residues

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§1.1 Contour Integrals and Tangents

A contour is a piecewise C^1 path

$$\gamma : [a, b] \to \mathbb{C}.$$

For a continuous function f on the image of γ, define

$$\int_\gamma f(z)\,dz = \int_a^b f(\gamma(t))\gamma'(t)\,dt.$$

The right-hand side is an ordinary complex-valued Riemann integral.

Let

$$a = t_0 < t_1 < \cdots < t_n = b$$

and choose $\tau_j \in [t_{j-1}, t_j]$. The Riemann sum is

$$\sum_{j=1}^{n} f(\gamma(\tau_j))(\gamma(t_j) - \gamma(t_{j-1})).$$

Each directed increment is multiplied by f, hence scaled and rotated before the increments are added. A contour integral is not an area under a graph; it is a limit of transformed tangent vectors.

The basic estimate follows immediately:

$$\left|\int_\gamma f(z)\,dz\right| \le \int_a^b |f(\gamma(t))|\,|\gamma'(t)|\,dt \le L(\gamma)\max_{z\in\gamma}|f(z)|,$$

where

$$L(\gamma) = \int_a^b |\gamma'(t)|\,dt.$$

This ML-estimate is the main error-control tool for contour deformation and limiting arguments.

§1.2 Orientation, concatenation, and reparametrization

The reverse contour is

$$\overline{\gamma}(t) = \gamma(a + b - t).$$

Substitution gives

$$\boxed{\int_{\overline{\gamma}} f(z)\,dz = -\int_\gamma f(z)\,dz.}$$

If $\gamma_1(b_1) = \gamma_2(a_2)$, the concatenated contour $\gamma_1 * \gamma_2$ traverses the first and then the second. Splitting the parameter interval yields

$$\boxed{\int_{\gamma_1 * \gamma_2} f(z)\,dz = \int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz.}$$

Let $\phi : [c, d] \to [a, b]$ be a piecewise C^1, increasing surjection with $\phi(c) = a$, $\phi(d) = b$. Then

$$\begin{aligned}\int_{\gamma\circ\phi} f(z)\,dz &= \int_c^d f(\gamma(\phi(s)))\gamma'(\phi(s))\phi'(s)\,ds \\ &= \int_a^b f(\gamma(t))\gamma'(t)\,dt.\end{aligned}$$

Thus the integral is unchanged by orientation-preserving reparametrization. If ϕ reverses orientation, the sign changes.

For the unit circle, all three parametrizations

$$e^{it}, \quad 0 \le t \le 2\pi; \qquad e^{2\pi it}, \quad 0 \le t \le 1; \qquad e^{2\pi it^5}, \quad 0 \le t \le 1$$

give the same oriented contour integral.

§1.3 The complex integral is the pullback of a one-form

Write

$$z = x + iy, \qquad f(z) = u(x, y) + iv(x, y), \qquad dz = dx + i\,dy.$$

Then

$$\begin{aligned}f(z)\,dz &= (u + iv)(dx + i\,dy) \\ &= (u\,dx - v\,dy) + i(v\,dx + u\,dy).\end{aligned}$$

Therefore

$$\int_\gamma f(z)\,dz = \int_\gamma (u\,dx - v\,dy) + i\int_\gamma (v\,dx + u\,dy).$$

A complex contour integral is two real line integrals coupled by the complex multiplication rule.

In differential-form notation, $f(z)\,dz$ is a complex-valued one-form. The parametrized formula is exactly pullback:

$$\int_\gamma f(z)\,dz = \int_{[a,b]} \gamma^*(f(z)\,dz).$$

This explains reparametrization invariance: integration is applied to the pulled-back one-form on the parameter interval.

§1.4 Primitives are equivalent to path independence

Let $D \subseteq \mathbb{C}$ be a domain and let $f : D \to \mathbb{C}$ be continuous.

Theorem 1.4.1

The following are equivalent.

(i) There is a holomorphic function $F : D \to \mathbb{C}$ with $F' = f$.

(ii) The integral of f depends only on the endpoints of the contour.

(iii) Every closed contour γ in D satisfies

$$\int_\gamma f(z)\,dz = 0.$$

Proof. If $F' = f$, the chain rule gives

$$\frac{d}{dt}F(\gamma(t)) = f(\gamma(t))\gamma'(t).$$

Hence

$$\int_\gamma f(z)\,dz = F(\gamma(b)) - F(\gamma(a)),$$

which proves (i)⇒(ii), and (ii)⇒(iii) is immediate.

Assume (iii). Fix $z_0 \in D$, and for $z \in D$ define

$$F(z) = \int_{\gamma_{z_0,z}} f(\zeta)\,d\zeta,$$

where $\gamma_{z_0,z}$ is any contour from z_0 to z. If two contours are chosen, following one and reversing the other gives a closed contour, so (iii) makes F well defined.

For small h, use the same contour to z and then the line segment from z to $z+h$:

$$\frac{F(z+h)-F(z)}{h} = \int_0^1 f(z+th)\,dt.$$

By continuity, the right-hand side tends to $f(z)$. Thus $F'(z) = f(z)$. □

The topology of the domain enters in condition (iii): local primitives may exist while a global primitive fails because closed loops can carry nonzero periods.

§1.5 The Cauchy–Goursat Theorem

Suppose $f = u + iv$ has continuous first partial derivatives on a region containing a positively oriented simple closed contour $\gamma = \partial\Omega$. Using the decomposition of $f\,dz$ and Green's theorem,

$$\begin{aligned}\operatorname{Re}\int_\gamma f(z)\,dz &= \int_\gamma u\,dx - v\,dy \\ &= \iint_\Omega (-v_x - u_y)\,dx\,dy,\end{aligned}$$

and

$$\begin{aligned}\operatorname{Im}\int_\gamma f(z)\,dz &= \int_\gamma v\,dx + u\,dy \\ &= \iint_\Omega (u_x - v_y)\,dx\,dy.\end{aligned}$$

The Cauchy–Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

make both integrands zero. Therefore

$$\int_\gamma f(z)\,dz = 0.$$

Goursat's theorem removes the assumption of continuous partial derivatives: complex differentiability alone is enough. On a simply connected domain, every closed contour is deformable through the domain to a point, and Cauchy–Goursat implies

$$\int_\gamma f(z)\,dz = 0.$$

By the preceding theorem, every holomorphic function on a simply connected domain has a primitive.

The conclusion fails when the domain contains a hole around a singularity. The model obstruction is $1/z$ on $\mathbb{C}^*$.

§1.6 Powers on a circle isolate one Fourier mode

For the circle traversed k times,

$$\gamma_k(t) = e^{ikt}, \qquad 0 \le t \le 2\pi,$$

one has

$$\begin{aligned}\int_{\gamma_k} z^m\,dz &= \int_0^{2\pi} e^{imkt} ike^{ikt}\,dt \\ &= ik\int_0^{2\pi} e^{ik(m+1)t}\,dt.\end{aligned}$$

If $m \neq -1$, the oscillation completes an integer number of periods and cancels. If $m = -1$, the integrand is the constant ik. Hence

$$\boxed{\int_{\gamma_k} z^m\,dz = \begin{cases} 2\pi ik, & m = -1, \\ 0, & m \neq -1. \end{cases}}$$

The exceptional exponent -1 is the coefficient that survives contour averaging. This is the local algebra behind residues.

§1.7 Winding number is the period of $dz/(z-a)$

Let γ be a closed contour avoiding a. Define

$$\mathrm{Ind}(\gamma, a) = \frac{1}{2\pi i}\int_\gamma \frac{dz}{z-a}.$$

For a circle centered at a traversed k times, the preceding computation gives

$$\mathrm{Ind}(\gamma, a) = k.$$

In general the index is an integer, equal to the endpoint change of a lifted argument of $\gamma(t) - a$.

It is invariant under homotopies that avoid a. For a smooth homotopy $\Gamma(t,s)$, differentiate under the integral sign:

$$\begin{aligned}\frac{d}{ds}\int_0^1 \frac{\partial_t\Gamma(t,s)}{\Gamma(t,s)-a}\,dt &= \int_0^1 \partial_t\left(\frac{\partial_s\Gamma(t,s)}{\Gamma(t,s)-a}\right)dt\\ &= 0,\end{aligned}$$

because the loop has the same initial and terminal point. Approximation or covering-space lifting gives the general continuous-homotopy statement.

Thus the integral depends on the homotopy class of the loop in $\mathbb{C}\setminus\{a\}$, not on its detailed shape.

§1.8 Closed need not mean exact on a punctured domain

On $\mathbb{R}^2\setminus\{0\}$, consider the real one-form

$$\alpha = \frac{1}{2\pi}\frac{-y\,dx + x\,dy}{x^2+y^2}.$$

A direct calculation gives

$$d\alpha = 0.$$

So α is closed. On the unit circle $x=\cos t$, $y=\sin t$,

$$\alpha = \frac{dt}{2\pi},$$

and therefore

$$\int_{S^1}\alpha = 1.$$

If $\alpha = dF$ were exact, every closed-loop integral would vanish. Hence α is closed but not exact.

This period represents the generator of

$$H^1_{\mathrm{dR}}(\mathbb{C}^*)\cong\mathbb{R}.$$

The complex form dz/z contains the same obstruction:

$$\frac{1}{2\pi i}\int_\gamma \frac{dz}{z} = \mathrm{Ind}(\gamma,0).$$

The failure of a global logarithm on $\mathbb{C}^*$ is the same phenomenon. Locally,

$$d(\log z) = \frac{dz}{z},$$

but no single-valued logarithm exists around a loop of nonzero winding.

§1.9 Recovering Values from the Boundary

Let f be holomorphic on a neighborhood of a positively oriented simple closed contour γ and its interior, and let z lie inside. Then

$$\boxed{f(z) = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{\zeta - z}\,d\zeta.}$$

To see the mechanism, write

$$\frac{f(\zeta)}{\zeta - z} = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}.$$

The first term has a removable singularity at $\zeta = z$, with value $f'(z)$, so its closed-contour integral vanishes. The second term contributes

$$f(z) \int_\gamma \frac{d\zeta}{\zeta - z} = 2\pi i \, f(z) \operatorname{Ind}(\gamma, z).$$

For a simple positively oriented contour containing z, the index is 1.

Differentiating under the integral sign gives

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z)^{n+1}} \, d\zeta.$$

For example,

$$\int_{|\zeta|=2} \frac{e^\zeta}{(\zeta - 1)^3} \, d\zeta = \frac{2\pi i}{2!} e = \pi i e.$$

§1.10 Residues and Global Integrals

If f has an isolated singularity at a, its Laurent expansion is

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - a)^n.$$

The residue is

$$\operatorname{Res}(f, a) = c_{-1}.$$

All other powers integrate to zero around a small circle, while $(z - a)^{-1}$ contributes $2\pi i$. Therefore

$$\int_{|z-a|=r} f(z) \, dz = 2\pi i \operatorname{Res}(f, a).$$

For a contour γ avoiding finitely many poles $a_1, \ldots, a_N$, the residue theorem is

$$\boxed{\int_\gamma f(z) \, dz = 2\pi i \sum_{j=1}^{N} \operatorname{Ind}(\gamma, a_j) \operatorname{Res}(f, a_j).}$$

Local Laurent coefficients are paired with global winding numbers.

Consider

$$f(z) = \frac{z + 1}{z(z - 1)}.$$

The residues are

$$\operatorname{Res}(f, 0) = -1, \qquad \operatorname{Res}(f, 1) = 2.$$

Hence

$$\int_{|z|=2} f(z) \, dz = 2\pi i(-1 + 2) = 2\pi i.$$

For the smaller contour $|z| = 1/2$, only the pole at 0 is enclosed, so

$$\int_{|z|=1/2} f(z) \, dz = -2\pi i.$$

The integrand has not changed; the answer changes because the winding data relative to the poles has changed.

§1.11 Counting zeros without solving a polynomial

Residues can count zeros, not only measure poles. Suppose p is a polynomial with no zero on a counterclockwise simple closed contour γ. Near a zero a of multiplicity m,

$$p(z) = (z-a)^m h(z), \qquad h(a) \neq 0, \qquad \frac{p'(z)}{p(z)} = \frac{m}{z-a} + \frac{h'(z)}{h(z)}.$$

The logarithmic derivative therefore has residue m at this zero. Applying the residue theorem gives the *argument principle*:

$$\boxed{\frac{1}{2\pi i}\int_\gamma \frac{p'(z)}{p(z)}\,dz = \#\{\text{zeros inside } \gamma\},}$$

counted with multiplicity. Equivalently, this number is the winding number of the *image curve* $p\circ\gamma$ about the origin. The integral is well defined only if no zero lies on the contour.

The striking step is that one can often count zeros without evaluating that integral. If g and h are holomorphic inside and on γ, and

$$\boxed{|h(z)| < |g(z)| \quad (z \in \gamma),}$$

then the curves $g(\gamma(t))$ and $(g+h)(\gamma(t))$ have the same winding number around zero. To see why, consider the continuous deformation

$$H_s(z) = g(z) + sh(z), \qquad 0 \le s \le 1.$$

On the boundary,

$$|H_s(z)| \ge |g(z)| - s|h(z)| > 0.$$

Throughout the deformation the image contour avoids zero, and winding number is integer-valued and cannot change continuously. The argument principle implies that g and $g+h$ have the same number of zeros inside. This is Rouché's theorem, derived directly from the homotopy invariance already used for contour integrals.

For a worked problem, consider

$$p(z) = z^4 + 3z + 1.$$

On the smaller circle $|z| = 1/2$, choose $g(z) = 3z$ and $h(z) = z^4 + 1$. Then

$$|g(z)| = \frac{3}{2}, \qquad |h(z)| \le 1 + \frac{1}{16} = \frac{17}{16} < \frac{3}{2}.$$

The function g has exactly one zero there, so p has exactly *one zero in* $|z| < 1/2$. On $|z| = 2$, choose $g(z) = z^4$ and $h(z) = 3z + 1$. This time

$$|g(z)| = 16, \qquad |h(z)| \le 7 < 16.$$

Therefore p has exactly *four zeros in* $|z| < 2$, namely all its zeros. Subtracting the counts proves that the intervening annulus

$$\boxed{\tfrac{1}{2} < |z| < 2}$$

contains exactly three zeros. No quartic formula was needed.

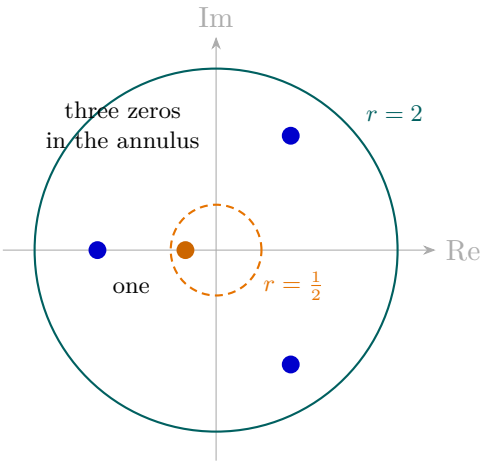


Figure 1.1: Approximate positions of the four zeros of $z^4 + 3z + 1$. The two root counts follow rigorously from boundary inequalities; the plotted positions are only a visual check.

We have followed a useful logical chain:

$$\begin{aligned} \text{contour integrals} &\longrightarrow \text{winding numbers} \\ &\longrightarrow \text{residues of } p'/p \\ &\longrightarrow \text{root counts stable under perturbation.} \end{aligned}$$

This is one reason analytic methods can tell us global facts about the solutions of algebraic equations without producing their individual roots.[1]

[1] For the argument principle and Rouché's theorem with contour examples, see Jeremy Orloff, MIT 18.04, Topic 11.